Introduction to Wave Mechanics

Physics 153 – Spring 2027

1. The Wave Function

Erwin Schrödinger was inspired by the success of de Broglie's matter wave hypothesis and initially dissatisfied with the abstract form of quantum mechanics, later called matrix mechanics, proposed by Heisenberg, Born, and Jordan. He invented what came to be called **wave mechanics** as an intuitive and visualizable approach. While the historical development involves advanced concepts from classical mechanics, we will not follow that path here.

In this course, we will describe the motion of a nonrelativistic particle of mass m using wave mechanics. At first, this may seem strange: why use waves to describe something we call a particle? Early on, de Broglie and even Schrödinger himself considered the possibility that a particle might literally be a wave spread out in space. However, this picture is not quite correct. When we measure the position of a particle, we always detect it at a definite location, like a localized object. On the other hand, experiments such as the single-particle double-slit experiment show that particles produce interference patterns, a hallmark of wave behavior, when many measurements are collected.

The modern resolution is subtle. A quantum system is described by a **state**, and in the *position representation* this state is given by a function $\psi(x)$, called the (position-space) wave function. **This function is not a physical wave in space.** Instead, it is a mathematical object that allows us to calculate probabilities—we call $\psi(x)$ the "probability amplitude." The quantity $|\psi(x)|^2$ gives the **probability density** for finding the particle at position x. The interference patterns seen in experiments arise from how the probability amplitudes combine, not from a particle physically spreading out like a classical wave. So rather than thinking of a particle as "sometimes a wave" it is more accurate to say: we are dealing with a system whose behavior is governed by a *wave function*, which encodes the probabilities of different measurement outcomes.

The probability to find the particle in the range $a < x < b$ is

$$P(a < x < b) = \int_a^b |\psi(x)|^2 dx$$

Since the total probability of finding a particle at all locations must be one, we say the wave function is normalized in the sense that it satisfies an integral relation of the form

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

A physical wave function is defined as one that can be normalized, and we normally require $\psi(x) \to 0$ for $x = \pm\infty$.

2. Measurements

The first major departure from classical physics is the role of **statistics**. In classical mechanics, if you know the state of a system precisely, you can in principle predict the outcome of any measurement with certainty. In quantum mechanics, even with complete knowledge of the system, individual measurement outcomes are generally **not deterministic**. Instead, the theory predicts **probabilities** for different possible results.

These probabilities are determined by the wave function. For example, $|\psi(x)|^2$ tells us the probability density for finding a particle at position x. Repeating the same experiment many times on identically prepared systems reveals a statistical pattern that agrees with these probabilities.

A second important feature of quantum mechanics is how physical quantities are represented. In quantum mechanics, quantities such as position, momentum, and energy are represented by **observables**, which are operators that "act on" the wave function. When we perform a measurement of some quantity, the only possible results are the **eigenvalues** of the corresponding operator. In other words, the allowed measurement outcomes are determined by solving an equation of the form

$$\hat{A}\,\psi = a\,\psi$$

where $\hat{A}$ is an operator, a is an eigenvalue, and ψ is an eigenfunction of the operator $\hat{A}$. The equation above is called an **eigenvalue equation.** It is a common misconception that any wave function can be an eigenfunction of any operator, and another misconception is that the eigenvalues are always discrete—they can be continuous. Lastly, the eigenvalue equation does not represent the measurement process—it simply tells you what values are possible for the observable.

This is another way in which quantum mechanics differs from classical physics. Instead of physical quantities having arbitrary values determined by initial conditions, the theory itself restricts the possible outcomes of measurements. In general, a system will not be in a single eigenstate of an operator. In that case, a measurement produces one of the allowed eigenvalues, with probabilities determined by the wave function. After the measurement, the system is found in the corresponding eigenstate—this is called "wave function collapse."

The probability to measure a particular *discrete eigenvalue* is given by the Born rule:

$$\text{prob}(a) = \left| \int_{-\infty}^{\infty} \psi_a^*(x)\psi(x)\,dx \right|^2$$

where $\psi_a(x)$ is the eigenfunction of $\hat{A}$ that satisfies $\hat{A}\,\psi_a = a\,\psi_a$ for precisely the eigenvalue a. If you're wondering where this all comes from, you'll have to take Phys 438A, or you can read my lecture notes online. For now, we will simply accept that it's part of a minimal set of postulates that allow us to explain observations and the results of experiments.

3. Schrödinger Equation

In the modern framework of quantum mechanics, between measurements, the state of a system is postulated to evolve over time in a very specific way that conserves total probability. Under the right circumstances, this leads to energy conservation in quantum mechanics. A nonrelativistic particle of mass m that is completely free of interactions has a classical energy given by

$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

where E is the energy of the particle and p is the x-component of momentum.

Electron diffraction and other experiments show that particles such as electrons produce interference patterns. This is a wave-like feature: something associated with the particle must have a wavelength and must be able to interfere. The de Broglie relations make this precise:

$$E = hf = \frac{h}{2\pi}\, 2\pi f = \hbar\omega$$

$$p = \frac{h}{\lambda} = \frac{h}{2\pi}\frac{2\pi}{\lambda} = \hbar k$$

where $\hbar$ is the reduced Planck's constant, ω is the angular frequency, and k is the wave number. In the context of quantum mechanics, there are only certain wave functions that will give the same energy and momentum of the particle each time those observables are measured, and they are given by complex plane wave solutions:

$$\psi(x,t) = Ae^{i(kx-\omega t)}$$

In the last section, we learned that physical quantities are represented by operators (observables), and the possible values obtained in measurements are given by the eigenvalues. Notice that we can obtain the momentum and energy of the plane wave by taking derivatives:

$$\frac{\partial}{\partial t}\psi(x,t) = \frac{\partial}{\partial t}Ae^{i(kx-\omega t)} = -i\omega\, Ae^{i(kx-\omega t)} = -i\frac{E}{\hbar}\,\psi(x,t)$$

$$\frac{\partial}{\partial x}\psi(x,t) = \frac{\partial}{\partial x}Ae^{i(kx-\omega t)} = ik\, Ae^{i(kx-\omega t)} = i\frac{p}{\hbar}\,\psi(x,t)$$

We can rearrange the expressions above to find eigenvalue equations for energy and momentum:

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = E\,\psi(x,t)$$

$$-i\hbar\frac{\partial}{\partial x}\psi(x,t) = p\,\psi(x,t)$$

Hence, the energy operator, also called the Hamiltonian $\widehat{H}$, is given by

$$\widehat{H} \doteq i\hbar \frac{\partial}{\partial t}$$

and the momentum operator is

$$\hat{p} \doteq -i\hbar \frac{\partial}{\partial x}$$

The dot is included above the equal sign to remind you that the operators are abstract entities, and we are writing down the way they act upon position-space wave functions only. In quantum mechanics, the energy operator is tied to changes in time while the momentum operator is tied to changes in position. Given the operators defined above, the eigenvalue equations become

$$\widehat{H}\,\psi(x,t) = E\,\psi(x,t)$$

$$\hat{p}\,\psi(x,t) = p\,\psi(x,t)$$

It is a common misconception that all wave functions satisfy the eigenvalue equations above. Only wave functions that describe a particle with definite energy and momentum will satisfy the corresponding eigenvalue equations! Not all wave functions are eigenfunctions.

In the words of Debye, "It is quite childish to speak of waves and not to have a wave equation." Let's substitute the energy and momentum operators into the total energy for a classical, nonrelativistic particle with mass m:

$$E = \frac{p^2}{2m} \rightarrow \widehat{H}\,\psi = \frac{\hat{p}^2}{2m}\,\psi$$

$$\left(i\hbar \frac{\partial}{\partial t}\right)\psi = \frac{1}{2m}\left(-i\hbar \frac{\partial}{\partial x}\right)^2 \psi = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\,\psi$$

The wave equation for a free particle is evidently given by

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi(x,t)}{\partial x^2}$$

Substituting the de Broglie relations into the energy expression, we also find

$$\hbar\omega = \frac{\hbar^2 k^2}{2m}$$

There are a few important remarks we can make before generalizing to the case of a non-isolated particle. Firstly, the complex plane wave solution indeed satisfies the free particle wave equation, but it's not the only solution. In fact, any linear combination of plane wave solutions will satisfy

the wave equation. The most general solution is given by a *wave packet*, which can be described using Fourier analysis (beyond the scope of this class). We'll return to superposition states later. Secondly, the relationship between ω and k is very different than the waves we studied earlier in the course. The angular frequency for particle waves is proportional to the square of the wave number, while for waves on a string ω is directly proportional to k.

Consider a particle that is interacting with a much larger sub-system whose dynamics are negligible compared to those of the particle. The interaction can be described by a potential energy function $U(x)$, and we modify the free particle wave equation to arrive at the Schrödinger equation for a nonrelativistic particle of mass m:

$$\boxed{i\hbar\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} + U(x)\psi(x,t)}$$

The Schrödinger equation is a linear partial differential equation with non-constant coefficients, and the solutions depend *sensitively* on the potential energy function. This model applies to a variety of physical systems whenever particle velocities are much smaller than the speed of light and *spin degrees of freedom* do not play a significant role. Physically, it describes a structureless particle with no relevant internal dynamics moving in a potential that does not couple to spin. This situation arises in several important contexts across atomic, nuclear, and condensed matter physics.

At this point, the ideas of quantum mechanics may feel abstract, unfamiliar, or only loosely motivated. **That reaction is normal.** We are intentionally approaching the subject in a limited way, focusing on the simplest cases and the most accessible tools. A deeper understanding requires more than definitions and equations. It comes from working through many examples, solving problems, and gradually building the necessary mathematical background. Concepts that seem unclear now often become much more concrete with practice. Quantum mechanics is not beyond understanding, but it does demand time and effort. The goal here is to provide a first pass; later, with more experience, the structure and logic of the theory may become clearer. For now, it is enough to become comfortable using these ideas, even if they do not yet feel completely intuitive.

4. Energy Eigenvalue Equation

When the potential energy is nonzero, we no longer expect to find solutions to the Schrödinger equation that are eigenstates of momentum—forces change momentum after all. We can almost always find solutions that have definite energy. In a definite-energy state, the energy of the particle will be the same value each time it's measured. Let $\psi_E(x,t)$ be a definite-energy state satisfying

$$\hat{H}\,\psi_E(x,t) = E\,\psi_E(x,t)$$

The wave function $\psi_E(x,t)$ is an energy eigenfunction. Substituting the energy operator (Hamiltonian), we find a first-order differential equation with constant coefficients:

$$i\hbar\frac{\partial\psi_E(x,t)}{\partial t} = E\,\psi_E(x,t)$$

By separation of variables (or using an integrating factor), we can easily solve this equation:

$$\psi_E(x,t) = \psi(x)e^{-iEt/\hbar} = \psi(x)e^{-i\omega t}$$

where $\omega = E/\hbar$ and $\psi(x)$ is a function that only depends on position. The first thing to note about the definite-energy state is that the probability density does not depend on time:

$$|\psi_E(x,t)|^2 = \psi_E^*(x)\psi_E(x) = \psi^*(x)e^{i\omega t}\psi(x)e^{-i\omega t} = |\psi(x)|^2$$

For this reason, definite-energy states are called stationary states. Substituting the definite-energy state into the Schrödinger equation, we find the **energy eigenvalue equation**, also called the time-independent Schrödinger equation:

$$\boxed{E\,\psi(x) = -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + U(x)\psi(x)}$$

where the factor of $e^{-i\omega t}$ cancels from both sides of the expression. It will be useful to write the eigenvalue equation in the form

$$\frac{d^2\psi}{dx^2} = -\frac{2m\big(E - U(x)\big)}{\hbar^2}\,\psi$$

where $\psi(x)$ is a time-independent wave function.

When the potential energy is constant, the energy eigenvalue equation reduces to a second-order linear differential equation with constant coefficients. Such equations are straightforward to solve, and because they are second order, the general solution necessarily contains two independent constants. These constants are not arbitrary; they are fixed by boundary conditions and matching conditions at interfaces between regions. It is important to understand both the form of these differential equations and the structure of their solutions. You should verify that the proposed solutions satisfy the equations by direct substitution and convince yourself that they represent the most general possible solutions.

When $E > U$, the solutions are oscillatory and the wavelength is proportional to $1/\sqrt{E-U}$:

$$\frac{d^2\psi}{dx^2} = -\frac{2m\big(E - U(x)\big)}{\hbar^2}\,\psi = -k^2\psi, \quad k \equiv \frac{\sqrt{2m(E-U)}}{\hbar} > 0$$

$$\text{Solution: } \psi(x) = A\,e^{ikx} + B\,e^{-ikx} = C\sin(kx) + D\cos(kx)$$

$$k = \frac{2\pi}{\lambda} \;\rightarrow\; \lambda = \frac{2\pi\hbar}{\sqrt{2m(E-U)}}$$

When $E < U$, the solutions are real exponential functions, and the decay length is proportional to $1/\sqrt{U-E}$:

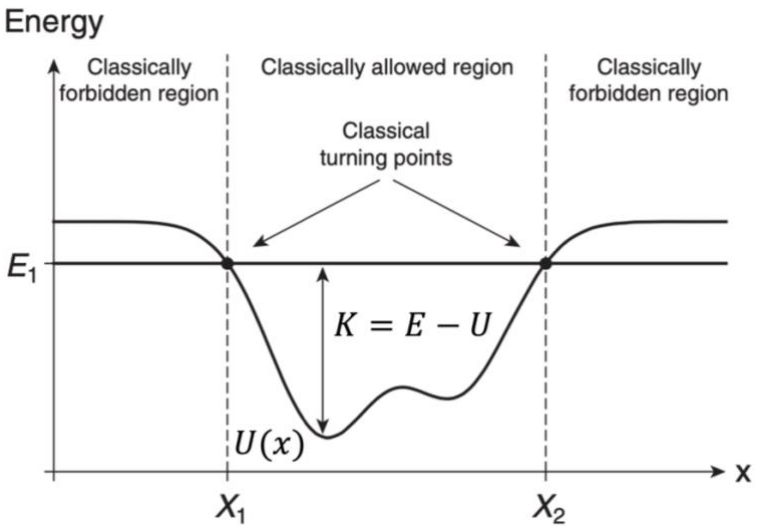
$$\frac{d^2\psi}{dx^2} = -\frac{2m(E-U)}{\hbar^2}\,\psi = q^2\psi, \quad q \equiv \frac{\sqrt{2m(U-E)}}{\hbar} > 0$$

$$\text{Solution: } \psi(x) = A\,e^{qx} + B\,e^{-qx} = C\sinh(qx) + D\cosh(qx)$$

$$q = \frac{1}{\ell} \;\rightarrow\; \ell = \frac{\hbar}{\sqrt{2m(U-E)}}$$

Most realistic potential functions are not constant, but many vary smoothly with position. When that is the case, you can think of the potential as being approximately constant over small regions, so the wave function in each small region behaves like it would for a constant potential. This gives us intuition; in regions where the particle's energy is higher than the potential, the wave tends to oscillate, while in regions where the energy is lower, it tends to decay. As the potential changes gradually, the wave function transitions smoothly between these behaviors. In this way, understanding the simple constant-potential cases provides a set of local building blocks that help you reason about solutions in almost any smoothly varying potential.

Energy Diagrams



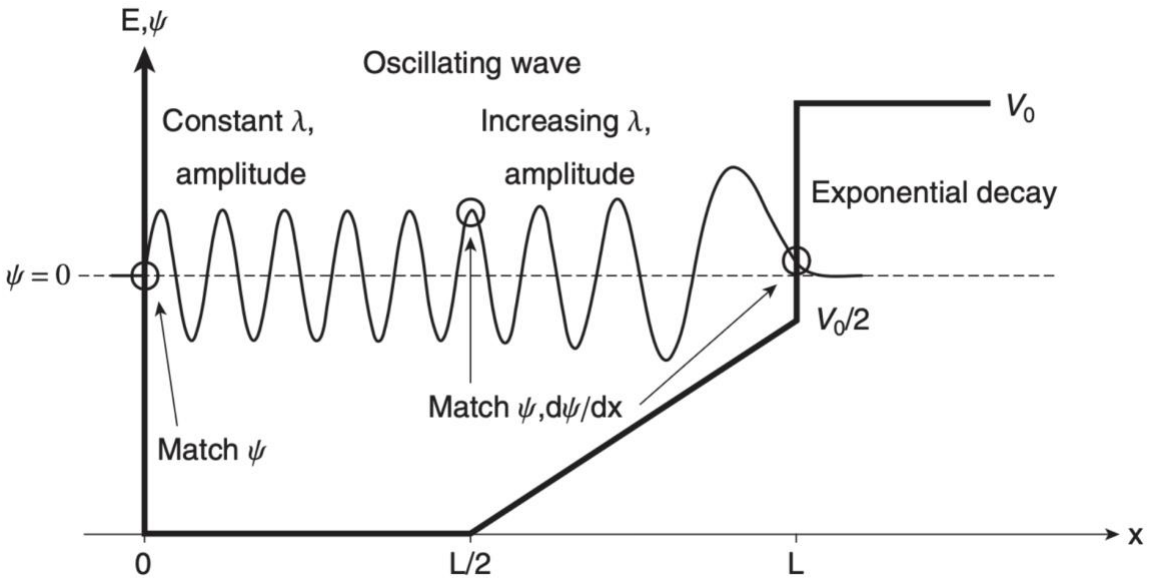
For a classical particle, the kinetic energy cannot be negative, so a classical particle with the energy E_1 chosen in the figure above has its motion constrained to the region between x_1 and x_2. These extreme points of the classical motion are called **classical turning points** and the region within the turning points is called the **classically allowed region**, while the regions beyond are called **classically forbidden regions**. Particles that have their motion constrained by the potential well are said to be in **bound states**. Particles with energies above the top of the potential well do not have their motion constrained and so are in **unbound states**. Note that the extent of the classically forbidden and allowed regions depends on the specific value of the energy, E_1, for a particular bound state.

Qualitative Solutions

Extending our results from earlier, we can make several general statements about the behavior of solutions to the energy eigenvalue equation for smoothly varying potentials:

1(a) In a classically allowed region, $E > U$, solutions are oscillatory,
1(b) the local wavelength is proportional to $1/\sqrt{E-U}$,
1(c) and the local amplitude is large for large wavelengths and small for small wavelengths.
2(a) In a classically forbidden region, $E < U$, solutions are real exponential functions,
2(b) and the local decay length is proportional to $1/\sqrt{U-E}$.
3(a) The time-independent wave function $\psi(x)$ must be continuous everywhere.
3(b) The first derivative of $\psi(x)$ must be continuous everywhere U is finite.

Rules 3(a) and 3(b) are called **matching conditions**, and they are necessary for the existence of the second derivative at all points (the energy eigenvalue equation is second order in position). We'll explain rule 1(c) in more detail later. Using these rules of thumb, we can get a very good idea of the wave function before we tackle the differential equation that gives us the exact solution in situations where the potential changes with position.



One final remark before we embark on solving the energy eigenvalue equation: the energy eigenstates are important because energy is a measurable and physically meaningful quantity, e.g., in spectroscopy experiments where discrete energy levels are observed. The corresponding solutions are therefore interesting in their own right, but they also play a broader role; they provide a set of "building blocks" for describing the system. The general solution can be written as a linear combination of energy eigenstates, and this decomposition gives useful insight into the system's dynamics and allows us to predict the outcomes of other measurements. We will not focus very much on the latter point, but it's important to keep in mind. Solving for the energy eigenstates is really just one step to solving a problem in quantum mechanics.

Amplitude Variation in a Potential Well

Recall 1(c), that in the classically allowed region, the amplitude is large for large wavelengths and small for small wavelengths. This can be understood as the result of the matching conditions 3(a) and 3(b). Consider a particle in a potential well (a classically allowed region bordered by large classically forbidden regions) for which there is a step change in the potential. To the left of the step, for $-a < x < 0$, the potential is 0, and for $0 < x < a$, the potential is a nonzero constant U_0. At all points in the range of the well $-a < x < a$, the energy of the particle is greater than U_0.

On the left side of the potential, a valid solution to the energy eigenvalue equation is given by

$$\psi_1(x) = A_1 \sin(k_1 x + \phi_1)$$

where A_1 and ϕ_1 are integration constants, and $k_1 = \sqrt{2mE}/\hbar = 2\pi/\lambda_1$. If you're keeping track, this is the third way to write the solution in the classically allowed region where the potential energy function is constant. The first derivative of the wave function is

$$\psi_1'(x) = k_1 A_1 \cos(k_1 x + \phi_1)$$

and it follows that

$$A_1^2 = \psi_1^2 + (k_1^{-1}\,\psi_1')^2 = \psi_1^2 + (\psi_1')^2(\lambda_1/2\pi)^2$$

The expression above holds to the right of the boundary as well (just repeat the analysis):

$$A_2^2 = \psi_2^2 + (k_2^{-1}\,\psi_2')^2 = \psi_2^2 + (\psi_2')^2(\lambda_2/2\pi)^2$$

where $k_2 = \sqrt{2m(E - U_0)}\,/\hbar = 2\pi/\lambda_2$. The expressions for A_1^2 and A_2^2 hold for all points in their respective domains, including the boundary at $x = 0$, where ψ and ψ' are continuous, i.e., $\psi_1(0) = \psi_2(0)$ and $\psi_1'(0) = \psi_2'(0)$. Since the potential is greater to the right of the boundary, $\lambda_2 > \lambda_1$, and it follows immediately that $A_2 > A_1$. Any smoothly varying potential can be approximated as a sequence of many small regions in which the potential is nearly constant, like a series of tiny step potentials, so the result derived here generalizes in a straightforward way.

There is an intuitive way to understand the change in amplitude. The probability density to the right of the boundary is given by the square of the wave function's amplitude, so a larger amplitude corresponds to a higher probability of finding the particle in that region. From a classical perspective, when the potential energy increases, the kinetic energy decreases, and the particle moves more slowly. A classical particle spends more time in regions where it is moving slowly, making it more likely to be found there. In a classically allowed region, the quantum result reflects the same trend; the probability density is larger where the potential energy is higher. This is one situation in which a quantum prediction aligns with classical intuition!

5. Superposition States

Now that we've made decent progress on understanding the Schrödinger equation and the possible solutions, let's return to the basics of "quantum logic." We'll do this through a concrete example involving momentum eigenstates. The momentum eigenvalue equation is

$$\hat{p}\,\psi_p(x) = p\,\psi_p(x) \;\Rightarrow\; -i\hbar\frac{\partial}{\partial x}\psi_p(x) = p\,\psi_p(x)$$

This is a first-order differential equation for the eigenfunction, and the general solution is

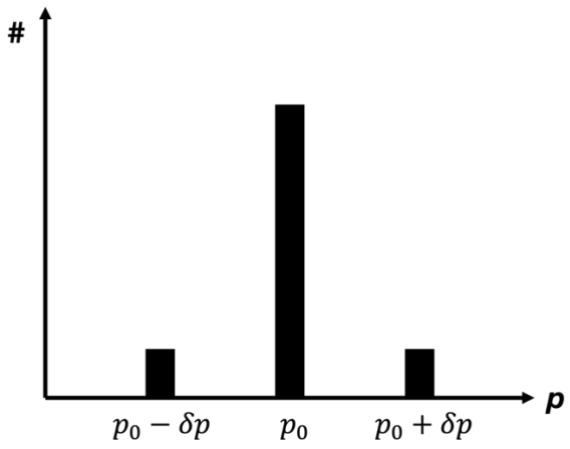
$$\psi_p(x) = Ae^{ikx}$$

where $p = \hbar k$. For a free particle, we already know this is an eigenstate of the energy operator, so the time-dependent wave function is

$$\psi_p(x,t) = Ae^{i(kx-\omega t)}$$

where $E = \hbar\omega = (\hbar k)^2/2m$. At any later time, if a measurement of momentum or energy is made on a system described by this wave function, the outcome will be precisely p or E respectively, every single time the measurement is made.

Quantum mechanics allows for states that encode *distributions* of outcomes. Given two solutions to the Schrödinger equation, a linear differential equation, their sum is also a solution. We call any linear combination of wave functions a **superposition state**. Consider an experiment in which repeated measurements of momentum on identically prepared systems yields the following distribution:



Clearly, p_0 is obtained more frequently than $p_0 \pm \delta p$ where $\delta p \ll p_0$. The wave function that gives this distribution of measurement outcomes can be written as

$$\psi(x) = A\left[\frac{1}{2}e^{i(k_0-\delta k)x} + e^{ik_0x} + \frac{1}{2}e^{i(k_0+\delta k)x}\right]$$

This wave function is a linear combination (superposition) of momentum eigenstates. For a free particle, each term is also an energy eigenstate, and the possible energies are

$$E_0 = \frac{p_0^2}{2m} = \hbar\omega_0$$

$$E_\pm = \frac{(p_0 \pm \delta p)^2}{2m} = E_0 \pm \frac{p_0}{m}\,\delta p = \hbar(\omega_0 \pm \delta\omega)$$

where we ignored the term involving δp^2 because it is negligible for small δp. The full time-dependent wave function is a superposition given by

$$\psi(x,t) = A\left[\frac{1}{2}e^{i[(k_0-\delta k)x-(\omega_0-\delta\omega)t]} + e^{i(k_0x-\omega_0t)} + \frac{1}{2}e^{i[(k_0+\delta k)x-(\omega_0+\delta\omega)t]}\right]$$

$$= Ae^{i(k_0x-\omega_0t)}\left[\frac{1}{2}e^{-i(\delta k\,x\,-\,\delta\omega\,t)} + 1 + \frac{1}{2}e^{i(\delta k\,x\,-\,\delta\omega\,t)}\right]$$

$$= A\exp\left[ik_0\left(x - \frac{\omega_0}{k_0}t\right)\right]\left\{1 + \cos\left[\delta k\left(x - \frac{\delta\omega}{\delta k}t\right)\right]\right\}$$

$$= 2A\exp\left[ik_0\left(x - \frac{\omega_0}{k_0}t\right)\right]\cos^2\left[\frac{\delta k}{2}\left(x - \frac{\delta\omega}{\delta k}t\right)\right]$$

In this form, we see that the first term is a complex traveling plane wave that moves with speed

$$\frac{\omega_0}{k_0} = \frac{\hbar\omega_0}{\hbar k_0} = \frac{p_0^2/2m}{p_0} = \frac{p_0}{2m}$$

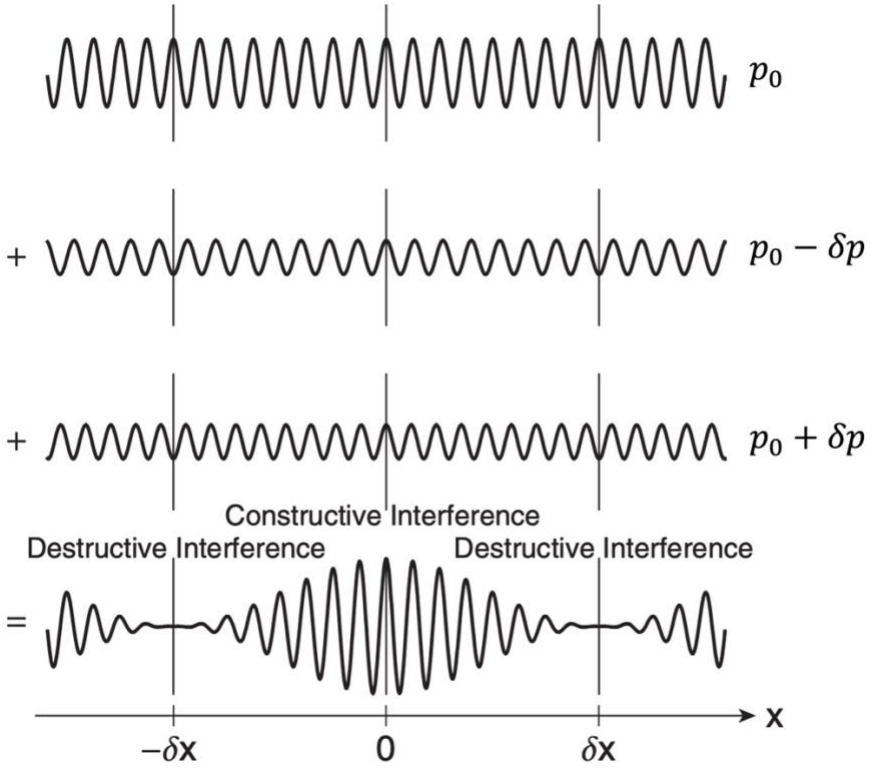
and the second term is a real-valued traveling wave that moves with speed

$$\frac{\delta\omega}{\delta k} = \frac{\hbar\delta\omega}{\hbar\delta k} = \frac{p_0\delta p/m}{\delta p} = \frac{p_0}{m}$$

The latter expression is the velocity of a classical particle with mass m and the former is *half* the classical velocity. The probability density function is given by $|\psi|^2$, so the complex exponential function will not contribute. The remaining term encodes *interference* between the momentum eigenstates. We observe locations where the eigenstates add constructively, and other locations where they add destructively. This is a relatively simple example of a *wave packet*. Let's write the superposition state in terms of momentum:

$$\psi(x,t) = 2A\exp\left[\frac{ip_0}{\hbar}\left(x - \frac{p_0}{2m}t\right)\right]\cos^2\left[\frac{\delta p}{2\hbar}\left(x - \frac{p_0}{m}t\right)\right]$$

We can visualize the wave packet and its components by (arbitrarily) taking the real part as in the graph below, which shows the eigenstates "before and after" taking their superposition:



This situation is analogous to the "beat phenomenon" that is well known for sound waves.

An important feature of a wave packet, and wave functions more generally, is the inverse relationship between the spread in momentum and the spread in position. We'll show through a simple argument that a wide distribution of momentum values is correlated with a narrow distribution of position values. Define δx to be the location of the first minimum in the envelope function for $x > 0$, where the cosine term equals 0. Since $\cos\theta = 0$ when $\theta = \pi/2$, at $t = 0$,

$$\frac{\delta p}{2\hbar}\delta x = \frac{\pi}{2} \;\Rightarrow\; \delta x\, \delta p = \pi\hbar$$

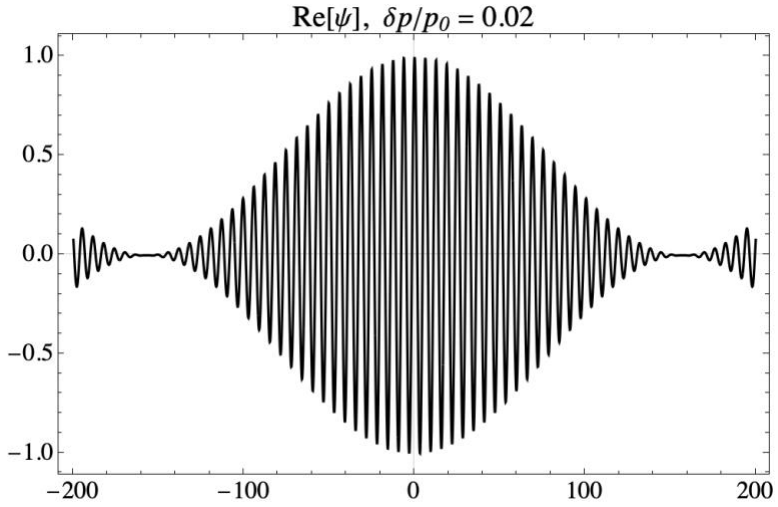
This relation is a quantitative expression of the concept explained earlier; as δp increases, we expect δx to decrease. It is common to refer to δp as the "uncertainty in the momentum" and δx as the "uncertainty in the position." The expression above is related to the famous uncertainty relation, which states the product of uncertainties in position and momentum satisfies

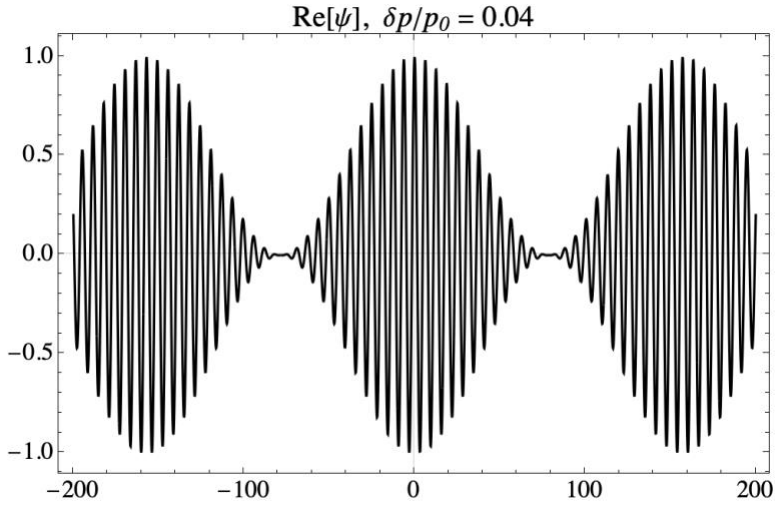
$$\Delta x\, \Delta p \geq \frac{\hbar}{2}$$

It is important to understand what the uncertainty relation does and does not imply. It is primarily a statement about the wave function, which encodes probability distributions for measurement

outcomes. A larger spread in momentum corresponds to a smaller spread in position, and vice versa. However, the uncertainty relation does not mean that individual measurements of position or momentum are inherently "fuzzy" or imprecise. Each measurement yields a definite value. The uncertainty refers to the spread in results obtained from many repeated measurements on identically prepared systems. Limitations in a single measurement may arise from experimental resolution, but that is a separate issue. Confusing these two ideas is a common mistake.

The graphs below compare two superposition states, one with a small spread in momentum and one with a larger spread in momentum. As the spread in momentum increases, observe the spread in position decreases correspondingly.

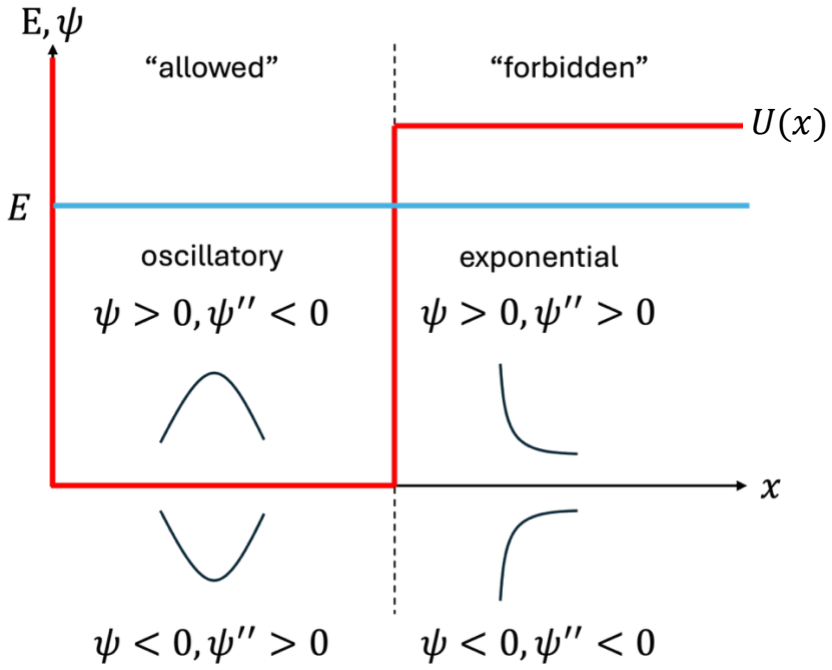




6. Bound States in Potential Wells

A **potential well** is a region of space where the potential energy is lower than in the surrounding regions, so a particle tends to be confined there. Classically, if the particle's total energy is less than the potential energy outside the well, it cannot escape and its motion is restricted to that region (it's in a bound state). In quantum mechanics, a particle in a potential well can still have a small probability of being found outside the well, but the well still largely confines the particle and leads to *discrete energy levels*. Quantization of energy is the result of imposing boundary conditions on the wave function.

Consider the potential well shown in the figure below:



By carefully considering the form of the energy eigenvalue equation in various regions, we can already get a good idea of what the energy eigenstates will look like. For instance, inside the well, the solutions will be oscillatory; the wave function curves toward the x axis, meaning it's concave down when $\psi > 0$ and concave up when $\psi < 0$. Outside the well, the solutions will exhibit exponential growth or decay, depending on the sign of the wave function. Convince yourself that these qualitative results follow directly from the energy eigenvalue equation:

$$\psi'' = -\frac{2m\big(E - U(x)\big)}{\hbar^2}\,\psi$$

In each of the examples that follow, start by drawing a qualitatively correct solution using the conditions discussed in section 4 where we introduced the energy eigenvalue equation. This will give you intuition, and since you will most likely be asked to sketch energy eigenfunctions on the final exam, it will also give you some practice!

Infinite Square Well

Consider a "super ball" bouncing between two perfectly elastic walls. We call this system a **particle in a box**. We observe three important characteristics of the classical system: (1) the ball flies freely between the walls, (2) the ball is reflected perfectly at each bounce, and (3) the ball remains in the box no matter how large its energy. These three observations are consistent with (1) zero force on the ball when it is between the walls, (2) infinite force on the ball at the walls, and (3) infinite potential energy outside the box.

The potential energy function for the classical super ball is a piecewise function given by

$$U(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & \text{otherwise}, \end{cases}$$

and we posit that the same function models the quantum mechanical super ball. We are interested in the definite-energy states of this system, and since any finite energy is less than the energy of the barriers, our definite-energy states will describe bound states. Inside the well, the solutions are clearly oscillatory. Outside the well, we may first consider a *finite square well,* where solutions exhibit exponential decay with decay length proportional to $1/\sqrt{U-E}$. In the limit of an *infinite square well,* the decay length tends to zero, and we conclude that ψ must be zero outside the well.

The energy eigenvalue equation inside the well is the same as that for a free particle:

$$\psi'' = -\frac{2mE}{\hbar^2}\psi = -k^2\psi, \quad k = \frac{\sqrt{2mE}}{\hbar}$$

and the general solution can be written in terms of constants A and B as

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

We now enforce the matching conditions (boundary conditions). The wave function must be continuous at the left and right boundaries of the well, i.e., at $x = 0$ and $x = a$. We do not, however, expect the first derivative to be continuous since the potential energy function is infinite at the boundaries. From continuity of the wave function at the left boundary,

$$x = 0: \quad \psi(0) = A\sin 0 + B\cos 0 = 0 \quad \Rightarrow \quad B = 0$$

and from continuity at the right boundary,

$$x = a: \quad \psi(a) = A\sin(ka) = 0$$

This is a transcendental equation that places limitations on the allowed values of the wave vector k. The wave vectors that satisfy this equation are

$$k_n = \frac{n\pi}{a}, \qquad n = 1, 2, 3, \ldots$$

where $n = 0$ is excluded because $\psi(0) = 0$ cannot be normalized and is therefore unphysical. Only *discrete wave vectors* are allowed, so this is called a **quantization condition**. The index n is the quantum number, which we use to label the quantized states and energies. You may recognize this quantization condition from our study of waves earlier in the course. We can rewrite the quantization condition in terms of the allowed wavelengths to find that

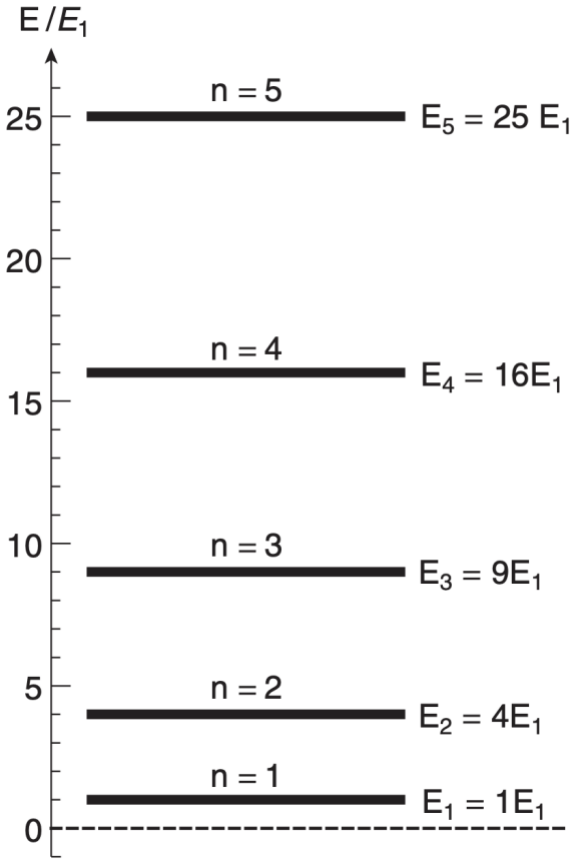
$$\lambda_n = \frac{2a}{n}$$

which is the same condition we found for standing waves on a string fixed at both ends! In words, the well must contain an integer number of half wavelengths.

Using the definition of the wave vector, we obtain an expression for the quantized energies:

$$\boxed{E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2ma^2}, \qquad n = 1, 2, 3, \ldots}$$

These allowed energies scale with the square of the quantum number n and produce the set of energy levels shown in the figure below. The ground state is the $n = 1$ level.



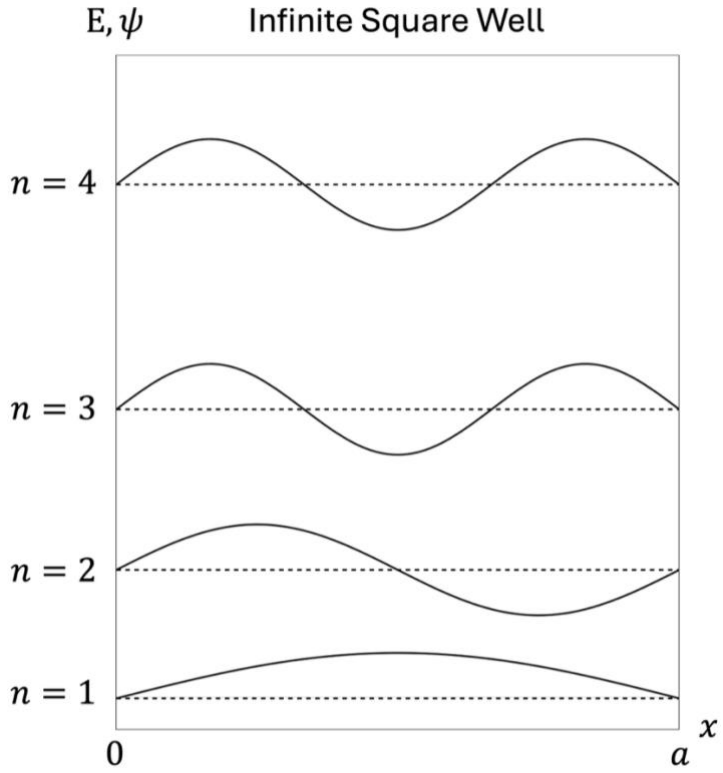
In order to find the constant A in the energy eigenfunctions, we need a third piece of information, which is that the wave function must be normalized to unity:

$$1 = \int_0^a |\psi(x)|^2 dx = |A|^2 \int_0^a \sin^2(k_n x)\, dx = |A|^2 \left(\frac{a}{2} - \frac{\sin(2ak_n)}{2} \right) = |A|^2 \cdot \frac{a}{2}$$

since $\sin(2ak_n) = \sin(2\pi n) = 0$. It follows that the energy eigenstates can be written as

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \qquad n = 1, 2, 3, \ldots}$$

Here are the first few energy eigenstates in the form of a unified schematic diagram:



Key properties of the energy eigenstates:

- The nth energy eigenstate has n antinodes and $n - 1$ nodes. This is a general characteristic of the energy eigenstates of any potential energy well.
- The allowed wave functions are either symmetric (even) or antisymmetric (odd) with respect to the center of the well. The symmetric states have "even parity" and the antisymmetric states have "odd parity." The parity alternates with successive energy levels. This is a general characteristic of symmetric potential wells.

Finite Square Well

Now let's make the problem a little more realistic by having the potential energy outside the well be finite instead of infinite. We still assume the well is square, which results in an infinite force at the walls. However, this new problem illustrates several important features of bound energy states that were not evident in the infinite well. A finite well can be used to model many real systems, such as an electron in a thin semiconductor.

In fact, for many years, the finite square well was found only in textbooks as an important steppingstone towards deeper understanding. Advances in semiconductor fabrication now allow engineers to create extremely thin layers of materials, forming structures called **quantum wells** that behave much like square potential wells. In these systems, electrons are confined to narrow regions, leading to discrete energy levels that can be tuned by adjusting the thickness or material properties. This control over energy levels is essential in modern devices such as semiconductor lasers, where quantum wells improve efficiency by enhancing specific light emissions.

The finite square well potential energy is written as

$$U(x) = \begin{cases} 0, & -a < x < a, \\ U_0, & \text{otherwise}, \end{cases}$$

where we have chosen a different position origin from the infinite well case for convenience. The finite square well will support bound states as long as the energy of the particle is less than the height of the barrier, i.e., bound states exist when $0 < E < U_0$.

Inside the well, the energy eigenvalue equation can be written as

$$\psi'' = -\frac{2mE}{\hbar^2}\psi = -k^2\psi, \quad k = \frac{\sqrt{2mE}}{\hbar} > 0$$

Outside the well, the energy eigenvalue equation can be written as

$$\psi'' = \frac{2m(U_0 - E)}{\hbar^2}\psi = q^2\psi, \quad q = \frac{\sqrt{2m(U_0 - E)}}{\hbar} > 0$$

The energy eigenvalue equation inside the box is identical to the one we solved for the infinite well potential. The differential equation outside the box is similar except the constant is positive instead of negative, giving real exponential solutions rather than complex exponentials.

We write the general solution as a piecewise function with solutions for each region:

$$\psi(x) = \begin{cases} A\, e^{qx} + B\, e^{-qx}, & x < -a \\ C\sin(kx) + D\cos(kx), & -a < x < a, \\ F\, e^{qx} + G\, e^{-qx}, & x > a \end{cases}$$

Notice that there are six undetermined constants in the general solutions, along with one unknown energy eigenvalue, for a total of seven unknowns. The matching conditions at the boundaries provide four equations, two from continuity of the wavefunction and two from continuity of its first derivative. The requirement that the wavefunction vanish as $x \to \pm\infty$ gives two additional conditions. Finally, normalization provides one more constraint, bringing the total to seven independent equations. The point of this counting argument is to show that the problem is, in principle, solvable before working through the details. Having as many independent equations as unknowns is a necessary (though not sufficient) condition for a well-defined solution.

We know the solutions will be even or odd, so let's focus on the even solutions for now. The even solutions that vanish at $x \to \pm\infty$ are given by

$$\psi_{\text{even}}(x) = \begin{cases} A\, e^{qx}, & x < -a \\ D\cos(kx), & -a < x < a, \\ Ae^{-qx}, & x > a \end{cases}$$

The boundary conditions at the right wall give

$$\begin{aligned} \psi_{\text{even}}(a): &\quad D\cos(ka) = Ae^{-qa} \\ \psi'_{\text{even}}(a): &\quad -kD\sin(ka) = -qAe^{-qa} \end{aligned}$$

We can find the energy condition by dividing the two equations, which yields

$$k\tan(ka) = q$$

Because both k and q are functions of the energy, this equation gives us a formula to find the allowed energies. It is a transcendental equation and must be solved numerically. For the odd solutions, a similar argument leads to the transcendental equation

$$-k\cot(ka) = q$$

A graphical solution for the allowed energies using these two transcendental equations is most useful here. There are many ways of doing this. One way involves defining a set of *natural units* where $\hbar = 2m = a = 1$. It may be uncomfortable at first, but learning to work with natural units is a valuable skill. The corresponding units of length, momentum, and energy are given by

$$[\text{length}] = a, \quad [\text{momentum}] = \frac{\hbar}{a}, \quad [\text{energy}] = \frac{\hbar}{2ma^2}$$

which can be inferred by dimensional analysis. In natural units, from the definitions of k and q,

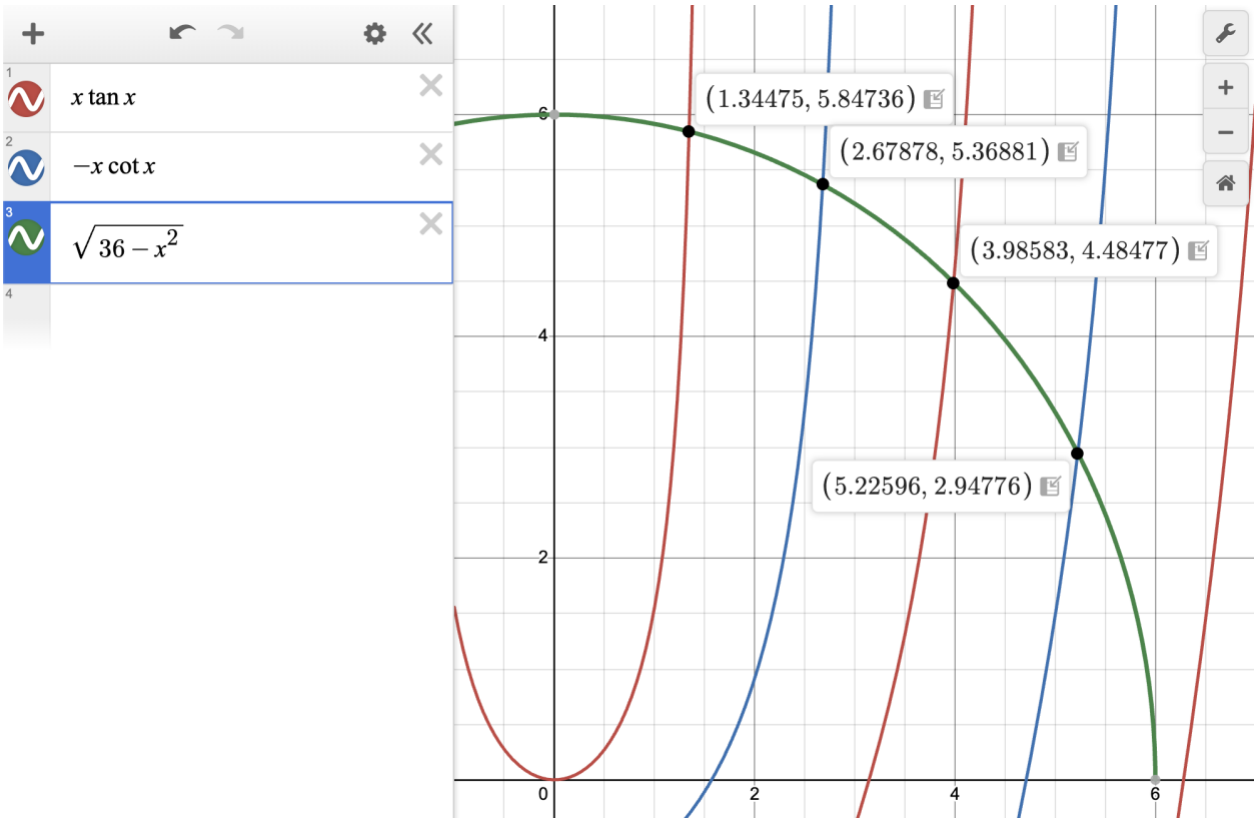
$$k^2 + q^2 = U_0 \Rightarrow q = \sqrt{U_0 - k^2}$$

$$\begin{aligned} \text{even solutions:} &\quad k\tan(k) = \sqrt{U_0 - k^2} \\ \text{odd solutions:} &\quad -k\cot(k) = \sqrt{U_0 - k^2} \end{aligned}$$

Let's consider an example where $U_0 = 36$ in natural units, which in regular units is the same as

$$U_0 = 36\,\frac{\hbar}{2ma^2}$$

so converting back is just a matter of multiplying by the energy unit. The allowed energies are found by plotting both sides of each equation and identifying their points of intersection. Here's an example from Desmos, which apparently can accomplish this quite easily:

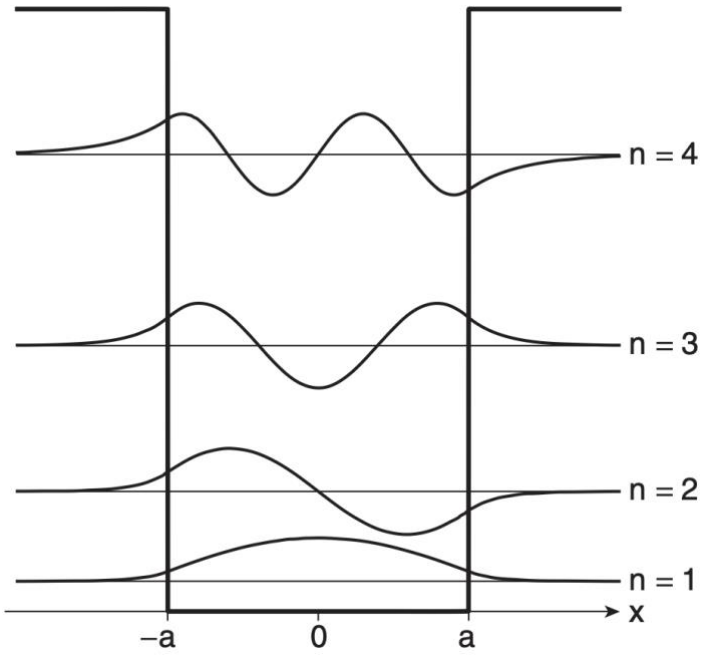


In natural units, $E = k^2$, so the allowed values of k and E for each of the four energy levels are

$$\begin{aligned} k_1 &= 1.35, \quad E_1 = 1.81 \\ k_2 &= 2.68, \quad E_2 = 7.18 \\ k_3 &= 3.99, \quad E_3 = 15.9 \\ k_4 &= 5.23, \quad E_4 = 27.3 \end{aligned}$$

The energy eigenstate wave functions are characterized by the allowed values of the parameters k and q. All that remains to do is normalize the wave function, which is straightforward but tedious.

We use a unified diagram to show the potential energy well, the allowed energies, and the allowed eigenstate wave functions superimposed in the figure below:



Note that the finite well eigenstates share many features with the infinite well states, with one major exception—they extend into the classically forbidden region. Quantum mechanical particles have a finite probability of being found where classical particles may not exist! This is a purely quantum mechanical effect and is commonly referred to as barrier penetration. The ability of the particle to penetrate the potential energy barrier leads to the phenomenon of tunneling.

Quantum Harmonic Oscillator

The quantum harmonic oscillator describes a particle moving in a potential that increases quadratically with position, which is a good approximation for many systems near stable equilibrium points (such as atoms in a molecule or lattice vibrations in a solid). Classically, the particle oscillates back and forth with a fixed frequency, but in quantum mechanics the behavior is different in several important ways.

Before we discuss the quantum harmonic oscillator, let's review a classical oscillator, such as a mass on a spring that obeys Hooke's law. The potential energy function is given by

$$V(x) = \frac{1}{2}kx^2$$

and the position of the oscillator over time can be described by the equation

$$x(t) = A\sin(\omega t - \phi)$$

where A and ϕ are integration constants that are determined by initial conditions, and ω is the *angular frequency* of the oscillator given by $\omega = \sqrt{k/m}$. In many situations, it is the angular frequency of the harmonic potential that is known, as opposed to the spring constant, so we replace

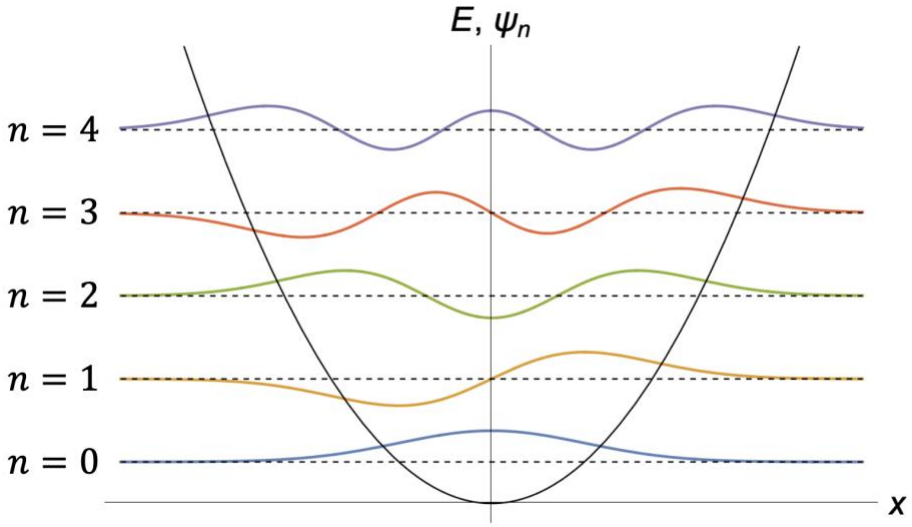
the spring constant with $k = m\omega^2$ in the potential energy function. It is beyond the scope of this class to solve the energy eigenvalue equation for the harmonic oscillator, but we can discuss the solutions and understand them using qualitative methods.

The allowed energies are quantized, meaning the particle can only have specific discrete values rather than a continuous range. These energies are given by

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \ldots$$

where there is a state with lowest possible energy, called the ground state, with $n = 0$ (rather than $n = 1$ as for the square wells). Notice that the energy levels for the harmonic oscillator are equally spaced, just as Planck proposed for the oscillators in the walls of the cavity that was used in the model for blackbody radiation!

The energy eigenfunctions are localized near the minimum of the potential, and they extend into classically forbidden regions where the particle would not be allowed in classical mechanics. As the energy increases, the wave functions spread out and begin to resemble classical motion more closely, oscillating in the allowed region and decaying outside it. Here are the first few eigenstates in a unified diagram:



Interestingly, in the limit of large n, the probability density $|\psi|^2$ begins to match the classical probability density, which predicts that a classical particle is more likely to be found near the turning points of the oscillator potential where the particle is moving more slowly.

In case you're interested, here are the energy eigenfunctions in terms of position:

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}\, x\right) e^{-m\omega x^2/2\hbar}$$

where H_n denotes the *Hermite polynomials*.

7. Scattering Theory

Unbound states have an energy that is greater than the potential energy at infinity, in contrast to bound states, which have an energy that is less than the potential energy at infinity. Bound states must "fit" into the potential well, which leads to energy quantization, while unbound states lie above the well and extend to infinity. A free particle is characterized by a wave packet, which is a superposition of plane wave solutions.

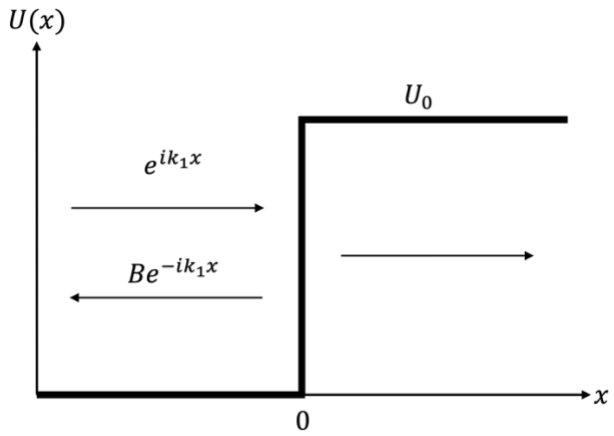
Consider a scenario in which a distant free particle travels to a region with a localized change in the potential energy function. The interaction will generally produce nonzero amplitudes for the wave packet to both **transmit** and **reflect**. If the wave packet is sharply peaked in momentum space so that its width in position space is large compared to that of the potential, we can treat the wave packet as a plane wave to good approximation. However, plane waves are stationary states for the free-particle Hamiltonian, so how can we calculate the probabilities for reflection and transmission? Think of the particles being continually projected onto a target; some incident flux is reflected, and some is transmitted. The probability of reflection will be the ratio of reflected flux to incident flux, and the probability of transmission will be the ratio of transmitted to incident flux.

In this class, we will only be concerned with finding the relative amplitudes for reflection and transmission; the larger the amplitude, the more likely it is for the particle to reflect or transmit through the region with a localized change in the potential energy function.

Counterintuitively, we find that particles have nonzero probability of *reflecting from potential wells* and *transmitting through potential barriers*!

Step Potential

Consider the potential energy diagram below:



Incident waves with definite energy and momentum arrive at the boundary and are partially reflected and transmitted. The energy of the particle is always greater than zero (the minimum of the potential energy) and the nature of the transmitted waves depends on the relationship between the energy and the height of the step potential. When $E > U_0$, the transmitted wave will be a plane

wave, and when $E < U_0$, the transmitted wave will decay exponentially into the barrier; we call this an **evanescent wave**. Let's investigate the first case, $E > U_0$.

The energy eigenvalue equation to the left of the step change is

$$\psi'' = -\frac{2mE}{\hbar^2}\psi = -k_1^2\psi, \quad k_1 = \frac{\sqrt{2mE}}{\hbar} > 0$$

and the energy eigenvalue equation to the right of the step change is

$$\psi'' = -\frac{2m(E-U_0)}{\hbar^2}\psi = -k_2^2\psi, \quad k_2 = \frac{\sqrt{2m(E-U_0)}}{\hbar} > 0$$

Since $k_2 < k_1$, we expect the wavelength to be longer to the right of the step change, and hence the amplitude of oscillations must be larger in that region as well. Let's confirm our intuition through a detailed analysis. The energy eigenfunction is a piecewise function given by

$$\psi(x) = \begin{cases} e^{ik_1x} + Be^{-ik_1x}, & x < 0 \\ Ce^{ik_2x}, & x > 0 \end{cases}$$

The matching conditions must be satisfied at $x = 0$, and they lead to the following relations:

$$\begin{aligned} \psi(0): &\quad 1 + B = C \\ \psi'(0): &\quad k_1(1 - B) = k_2C \end{aligned}$$

These are simultaneous equations for B and C, which can be solved to give the amplitudes of the reflected and transmitted waves respectively:

$$B = \frac{k_1 - k_2}{k_1 + k_2}$$

$$C = \frac{2k_1}{k_1 + k_2}$$

For the step potential we are considering here, since $k_2 < k_1$, we find $C > B$, just as we predicted. If the step becomes very small, $k_2 \to k_1$, $B \to 0$, and $C \to 1$, so there is complete transmission.

One interesting case exposes a limitation in this analysis. If the barrier becomes very large but still finite, $k_2 \ll k_1$, and it follows that $B \to 1$ while $C \to 2$. How can there be complete reflection of the wave while there is also a nonzero transmission? The missing ingredient is the *speed of the particles in the transmitted region.* The number of particles that are transmitting per unit time vanishes in the region $x > 0$ due to the vanishing speed of the particles in that region. As the barrier becomes large, the probability of reflection approaches 1 while that of transmission approaches 0.